A Graph-Attention Surrogate for Solver-Free Multi-Objective Inverse Design of Pixelated Metasurfaces

Saarthak Gupta, Anand Kumar, M. Gopal Krishna, and Debdeep Sarkar

***Abstract*—Optimizing a pixelated metasurface with a full-wave solver is slow, because every candidate layout needs its own simulation. This paper presents a neural-network surrogate that predicts the complex co- and cross-polarized reflection of a 16-by-16 pixel layout from 2 to 12 GHz in about a tenth of a millisecond. It combines a convolutional backbone with pixel-wise graph attention, learned frequency queries and a bidirectional recurrent layer, and keeps the reflected power below unity. Trained on each layout class of a public dataset with one architecture, it predicts the co-polarized reflection of shape combinations and random layouts with a 68–76 % lower error than the class-specific reference networks. Driven by the surrogate, a multi-objective particle swarm optimizer designs three reflective devices from random pixel layouts, each for bandwidth, polarization quality or reduction depth, and few corner contacts and isolated pixels. Scoring the bandwidth symmetrically about the design frequency keeps that frequency away from the band edges. Every selected design is checked with two full-wave solvers, a finite-difference time-domain code and CST Studio Suite. All 12 linear-to-circular polarization converters meet their target in both, with 13.2–18.4 % 3-dB axial-ratio bandwidth at 10 GHz, and so do all 12 polarization rotators, with 13.9–16.8 % bandwidth of over 90 % polarization conversion at 10.2 GHz; none of the $111{,}000$ dataset layouts meets either target. A coding checkerboard of a cell and its mirror image reduces the specular reflection by more than 10 dB over a 16.0 % band for both polarizations with tiles of six by six cells.**

***Index Terms*—Metasurfaces, inverse design, surrogate modeling, particle swarm optimization, finite-difference time-domain (FDTD) method.**

I. Introduction

METASURFACES, thin periodic arrays of sub-wavelength scatterers, control the amplitude, phase and polarization of reflected and transmitted waves [1]. A pixelated metasurface divides the unit cell into an $N \times N$ grid of small metal patches, each either present or absent [12], [13]. Patches, rings, crosses and split rings are all special cases of such a grid, and the grid can be printed directly on a circuit board. The number of possible layouts, $2^{N^2} = 2^{256}$ for a 16×16 grid, is far too large to try one by one.

Pixelated layouts are therefore found by optimization. A genetic algorithm [4] or a particle swarm [5], [6] proposes layouts, a full-wave solver computes their response, and the optimizer uses the result to propose better layouts [3]. The solver, not the optimizer, is the cost: a swarm of 100 particles over 100 generations needs 10^4 simulations, and multi-objective problems, whose result is a set of trade-off designs (a Pareto front) [7], need more.

A surrogate model is a neural network trained on simulated data to predict the response of a layout [3]; once trained, it answers in milliseconds and can replace the solver inside the loop. For cells described by a few dimensions this is well established [8], [9], and generative networks have proposed new shapes [10], [11]. Free-form pixel layouts have been designed in this way [12], but they are harder to learn, because the response depends on how the pixels connect over the whole cell. The reference study is the metasurface design deep convolutional neural network (MSDCNN) [13], which led to the public SUTD-PRCM dataset [14]: it needed a different ResNet [15] for different layout classes and predicted random layouts least accurately. A convolutional network sees only a small neighbourhood per layer and must stack many layers before it can relate distant parts of the layout, which decide current paths and resonances.

The proposed surrogate extends a convolutional network in two ways. Pixel-wise graph attention (PGA) [16], related to graph attention [17], lets each pixel weight its eight neighbours according to the layout, so that connected metal can be followed explicitly. Learned frequency queries [18], [19] let each output frequency pick out the parts of the layout that matter for it. The surrogate predicts both the co-polarized reflection r_{xx} and the cross-polarized reflection r_{yx}, which polarization-converting and coding surfaces need, and it is used to design three such devices without a solver in the loop. The designs are checked with a full-wave solver.

The main contributions are as follows.

1) A compact surrogate with graph attention and a passive two-output layer, one architecture for all four SUTD-PRCM classes, more accurate than the class-specific MSDCNN networks.
2) A rule for the latent size of a VAE of random layouts, below which the VAE ignores its input.
3) Solver-free multi-objective design of three devices: a linear-to-circular (LP-to-CP) polarization converter, a 90° polarization rotator and a 1-bit coding checkerboard for radar cross-section (RCS) reduction, with a band objective that keeps the design frequency away from the band edges.

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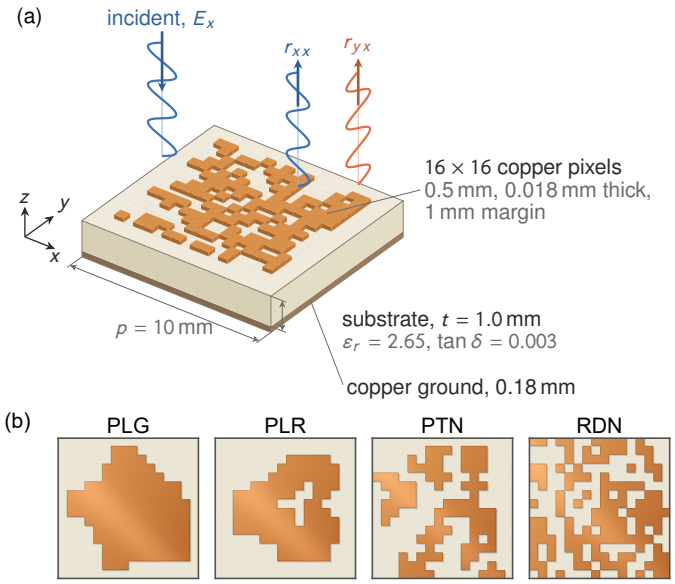


Fig. 1. (a) SUTD-PRCM unit cell (layer thicknesses not to scale); an x-polarized plane wave is reflected into r_{xx} and r_{yx}. (b) One layout of each class.

4) Verification of every selected design with an FDTD solver validated against the dataset and with CST, including periodic checkerboards of 2×2 tiles, which show how large the tiles of a coding surface must be.

Section II describes the dataset, Section III earlier surrogates, Section IV the surrogate, Section V its accuracy, Section VI the VAE, Section VII the design method, Section VIII the full-wave verification, and Section IX the designs.

II. Dataset and Unit Cell

The SUTD-PRCM unit cell [14] (Fig. 1) is a 16×16 grid of copper pixels, each 0.5×0.5 mm and 0.018 mm thick, inside a 1 mm margin, giving a 10 mm square cell. The pixels lie on a substrate with $\varepsilon_r = 2.65$, $\tan\delta = 0.003$ and thickness 1.0 mm, backed by a 0.18 mm copper ground plane. The cell is repeated periodically and illuminated at normal incidence by a plane wave polarized along x; no power is transmitted. The dataset stores, for every layout, the complex reflection coefficients r_{xx} (reflected x-polarized field over the incident field) and r_{yx} (reflected y-polarized field) from 2 to 12 GHz at 1001 frequencies, computed in CST. A layout is a 16×16 binary matrix with 1 for copper; rows run along x, the incident electric field.

The four classes are filled polygons (PLG, 30,000 layouts), polygon rings (PLR, 60,000), combinations of simple shapes (PTN, 60,000) and random pixels (RDN, 111,000), in which every pixel is copper with probability one half. RDN contains every possible layout and is the hardest class for a surrogate; it is also the class used for design. As in [18], every tenth frequency is kept, i.e., $K = 101$ frequencies 0.1 GHz apart. Following MSDCNN [13], each class network is trained on 27,000 layouts of its class and tested on 3,000 others, and a second RDN network is trained on 108,000 RDN layouts. Networks are named after their training set (e.g., PTN27000, RDN108000) and tested on the 3,000 test layouts of each class, which no network sees in training; no validation set is used.

III. Surrogate Models in Earlier Work

Table I lists surrogate models for metasurface and antenna design; reviews are given in [3], [30]. For parametric design spaces (a), networks map a few dimensions to a spectrum and are used inside an optimizer [21] or with an inverse network [8], [20]; a few thousand simulations suffice, but the shapes are fixed in advance. For pixel design spaces (b)–(d), forward CNNs predict a single-frequency phase [22] or $|S_{11}|$ of 12×12 pixel antennas [23], [24] and are used with lookup tables, binary swarms or inverse networks; generative models [25], [26] return one layout per target response; graph networks model coupling in large arrays [27], [28]. Each study uses its own dataset, often reduced by symmetry to 8×8 free pixels, so accuracies cannot be compared. SUTD-PRCM (e) is the only public set of free-form 16×16 layouts with four classes, and three surrogates have been reported on it: MSDCNN [13], a network architecture search [14], [29] that locates resonances, and the edge-aware surrogate [18], tested on a 30,000-layout subset of the dataset with a shallow CNN encoder.

The designs of Section IX place three demands on the surrogate that no entry of Table I meets together: it must predict both the complex co- and cross-polarized reflection at every frequency of the band, from which axial ratio, polarization conversion and the reflection of a checkerboard follow; it must follow how metal connects across free-form random layouts; and it must be fast, because the swarm scores 10^4–10^5 layouts per search against three objectives, which a generative model returning one layout per target cannot do.

IV. Surrogate Model

The surrogate maps a layout x to the complex reflections $\hat{S}_{xx}(f_k)$ and $\hat{S}_{yx}(f_k)$ at the 101 frequencies (Fig. 2): a convolutional backbone with graph-attention blocks turns the layout into 16 feature vectors, frequency cross-attention and a bidirectional GRU turn them into one vector per frequency, and an output layer turns each vector into two reflection coefficients.

A. Layout Encoder

A 3×3 convolution with 64 channels, batch normalization and $\tanh$ keeps the 16×16 resolution. Three stages of residual blocks [15] follow, with 3, 4 and 3 blocks and 64, 128 and 256 channels. A block, as in MSDCNN [13], adds a residual branch $\mathcal{F}$ (3×3 convolution, batch normalization, LeakyReLU with slope 0.2, 3×3 convolution, batch normalization) to a shortcut $\mathcal{S}$ and applies a ReLU, $\mathbf{y} = \mathrm{ReLU}\big(\mathcal{F}(\mathbf{u}) + \mathcal{S}(\mathbf{u})\big)$; the shortcut lets the gradient bypass the convolutions, so many blocks can be stacked [15]. The first block of stages 2 and 3 has stride 2 and a shortcut of a strided 1×1 convolution with batch normalization, giving 16×16, 8×8 and 4×4 maps. Each stage is followed by three PGA blocks. After the backbone, channel dropout (0.2) is applied in training, and the 16 positions of the 4×4 map become 16 tokens of 256 numbers, each describing one region of the layout and its surroundings.

A PGA block [16] lets each position of a feature map F (C channels, $N = HW$ positions) collect information from

TABLE I
SURROGATE MODELS FOR METASURFACE AND ANTENNA DESIGN

Work	Design space	Network	Predicted output	Used with
(a) Parametric design spaces				
Liu [8], 2018	thin-film stack (thicknesses)	fully connected, tandem	transmission spectrum	inverse network
Niu [20], 2026	1-bit cell (parameters)	tandem + deep Q-network	absorption, polarization conversion	array arrangement
Yu [21], 2026	square-loop FSS (dimensions)	graph attention + LSTM	$\lvert S_{21}\rvert$ spectrum	VAE-PSO
(b) Pixel design spaces: forward surrogates (own datasets)				
Zhu [22], 2021	16×16 coding cell, 8×8 free	Inception-V3, transfer learning	reflection phase, one frequency	phase-to-pattern lookup
Gupta [23], [24], 2023–25	12×12 pixel antenna	CNN + tandem inverse; transfer learning	$\lvert S_{11}\rvert$	inverse network; binary PSO
(c) Pixel design spaces: generative (inverse) models				
Niu [25], 2023	three 128×128 layers	diffusion model	layout from target S-parameters	–
Lyu [26], 2026	multilayer pixelated loop FSS	multichannel conditional GAN	layout from target filter specifications	–
(d) Graph networks for large arrays				
Khoram [27], 2023; Bao [28], 2025	arrays of meta-atoms	graph neural network; graph attention	near field; element response	large arrays; metalens and hologram design
(e) SUTD-PRCM: public free-form 16×16 dataset, four classes, 2–12 GHz				
MSDCNN [13], 2021	PLG, PTN, RDN; one network per class	ResNet18S / ResNet34S	$\lvert r_{xx}\rvert$, 32 frequencies	inverse network
NAS [14], [29], 2022	all four classes	searched shallow, wide CNN	resonance locations, $\lvert r_{xx}\rvert$, $\lvert r_{yx}\rvert$	–
Edge-aware [18], 2026	30,000-layout subset	shallow CNN + cross-attention + BiGRU	complex r_{xx}, 101 frequencies	–
This work	**all four classes, one architecture**	**PGA-ResNet + cross-attention + BiGRU**	**complex r_{xx} and r_{yx}, 101 freq.**	**MOIPSO, 3 objectives, 3 devices**

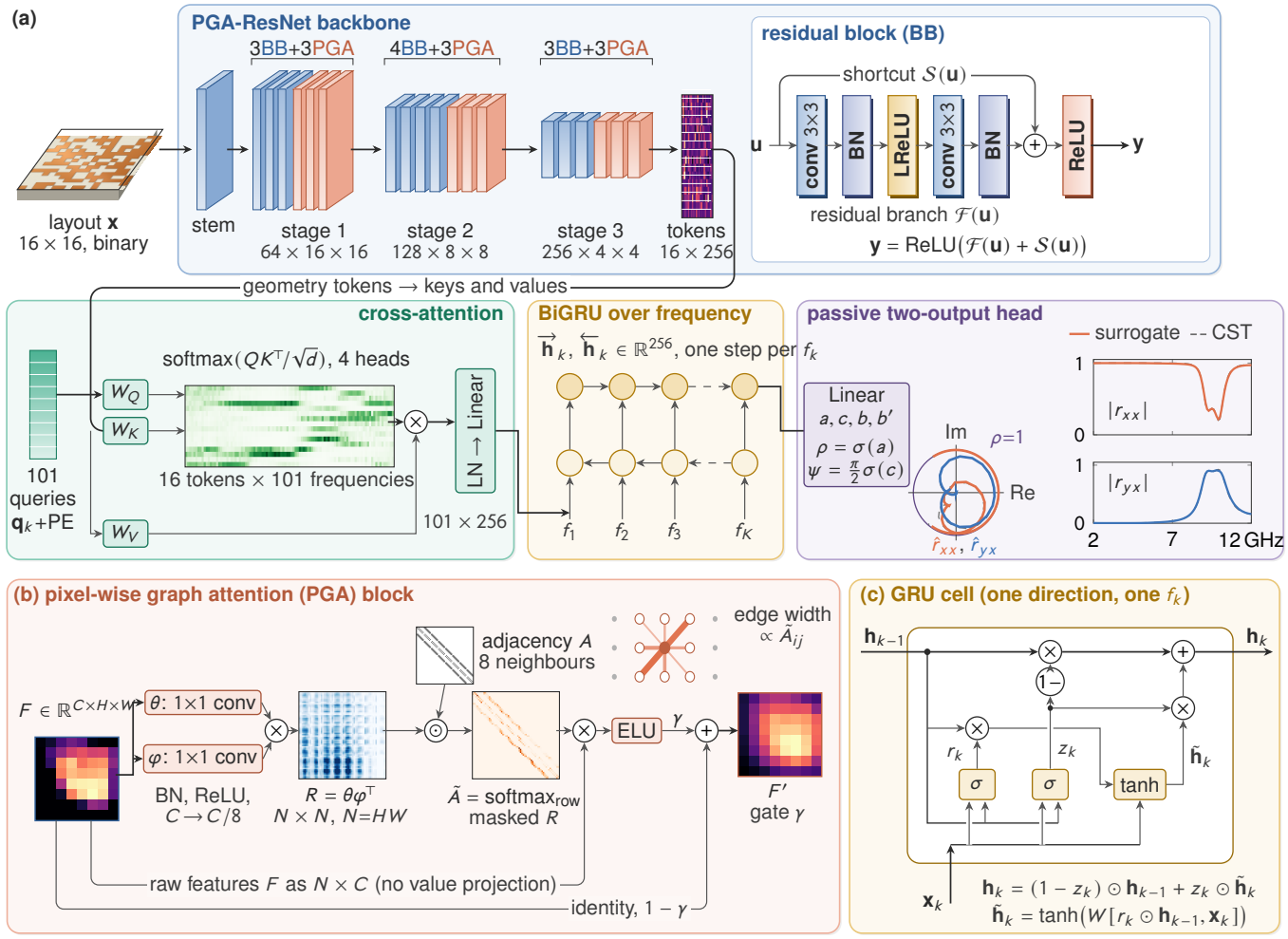


Fig. 2. Surrogate model. (a) Backbone: a 3×3 stem convolution and three stages of residual blocks (BB, blue) and PGA blocks (orange); block height shows the feature-map size and width the number of channels. Top right: one residual block; its shortcut $\mathcal{S}$ is the identity, or a 1×1 convolution with BN and stride 2 at the start of a stage. Below: frequency cross-attention, bidirectional GRU and the passive two-output layer (5), with the attention weights, token values and spectra of the trained two-output RDN108000 for the test layout shown (dashed: CST). (b) PGA block. (c) GRU cell.

its eight neighbours with layout-dependent weights (Fig. 2(b)). The neighbour graph $A_{ij} = 1$ if j is one of the eight neighbours of i depends only on the grid. Two 1×1 convolutions θ, φ with batch normalization and ReLU reduce C to $C/8$

and give scores $R_{ij} = \theta(F)_i^\top \varphi(F)_j$; a softmax over the neighbours gives the weights

$$\tilde{A}_{ij} = \frac{A_{ij} \exp R_{ij}}{\sum_{j'} A_{ij'} \exp R_{ij'}}. \quad (1)$$

Each position receives the weighted sum of its neighbours' original features, $\tilde{A}F$ (no value projection: θ and φ only decide how much each neighbour contributes), which passes an ELU and is mixed with the input through a learned scalar,

$$F' = (1-\gamma)\, F + \gamma\, \mathrm{ELU}\big(\tilde{A}F\big), \quad (2)$$

with $\gamma = 0$ at the start, so each block first passes its input unchanged; after training, $|\gamma|$ lies between 0.15 and 0.39 in every block of RDN108000. Unlike a convolution, whose weights are the same for every neighbourhood, a PGA block can weight a neighbour that continues a metal path more than one across a gap, and metal connectivity decides the current paths and resonances. The nine blocks add 65,193 parameters.

B. Frequency Decoder and Output Layer

Following [18], each frequency f_k has a learned query q_k of 256 numbers plus a fixed sinusoidal code of the frequency index [19], and collects information from the 16 tokens T (after layer normalization) by four-head attention,

$$\mathrm{Attn}(q_k, T) = \mathrm{softmax}\left(\frac{(W^Q q_k)(W^K T)^\top}{\sqrt{d}}\right) W^V T, \quad (3)$$

where W^Q, W^K, W^V are learned projections and $d = 64$ is the size of each head, followed by layer normalization and a linear layer. The 101 vectors $\mathbf{x}_k$ then pass in frequency order through a bidirectional GRU [31] with 256 units per direction,

$$\mathbf{h}_k = (1 - z_k) \odot \mathbf{h}_{k-1} + z_k \odot \tanh\big(W[r_k \odot \mathbf{h}_{k-1}, \mathbf{x}_k]\big), \quad (4)$$

with update and reset gates z_k, r_k, sigmoid functions of $\mathbf{h}_{k-1}$ and $\mathbf{x}_k$; one direction runs up and one down in frequency, so each prediction knows the response on both sides, which keeps spectra smooth and resonances consistent.

After dropout (0.2), a linear layer maps each frequency's 512 numbers to four numbers (a_k, c_k, b_k, b'_k), which give

$$\hat{S}_{xx} = \rho_k \cos\psi_k\, e^{jb_k}, \quad \hat{S}_{yx} = \rho_k \sin\psi_k\, e^{jb'_k}, \quad (5)$$

with $\rho_k = \sigma(a_k)$ and $\psi_k = \frac{\pi}{2}\sigma(c_k)$, where σ is the sigmoid. The total reflected power is $|\hat{S}_{xx}|^2 + |\hat{S}_{yx}|^2 = \rho_k^2 < 1$ by construction: the predicted surface never reflects more power than it receives, whatever layout the optimizer proposes. (A separate sigmoid per output would allow a total of up to 2.) The network has 5.83 M parameters.

C. Loss and Training

The physics-guided loss of [18] is applied to each output and summed,

$$\mathcal{L} = \mathcal{L}_{\mathrm{reim}} + 0.5\,\mathcal{L}_{\mathrm{dB}} + 0.3\,\mathcal{L}_{\mathrm{lin}} + 0.3\,\mathcal{L}_{\mathrm{ph}} + 0.08\,\mathcal{L}_{\mathrm{pass}} + 0.06\,\mathcal{L}_{\mathrm{TV}} + 0.02\,\mathcal{L}_{\mathrm{curv}} + 0.02\,\mathcal{L}_{\mathrm{KK}}, \quad (6)$$

with the terms of Table II. $\mathcal{L}_{\mathrm{reim}}$ (weight 1) compares the complex values. The magnitude is also compared in decibels, which makes the network place resonance dips at the right frequency and depth, and in linear scale, which weights the near-total reflection that covers most of the band. The phase term compares unit phasors and so has no $\pm 180°$ jumps. For r_{yx}, whose magnitude is often near zero, the decibel floor is raised to -40 dB and the phase term is weighted by $\min(1, |S|/0.1)$. Of all r_{yx} values, 33 % (RDN) to 62 % (PLG, PLR, PTN) lie below -40 dB, where they carry 10^{-4} of the incident power and change none of the quantities of Section VII; without the floor, the decibel term would count a difference between -80 and -60 dB like a misplaced resonance. For a small phase error $\Delta\phi$ the complex error is about $|S|\Delta\phi$, so the weight makes the phase term count in proportion to the error it causes, and every r_{yx} above -20 dB counts fully. Trained without these two settings, the networks have a 12–41 % higher $|r_{xx}|$ error on PTN and RDN and a 2.6–2.7 times higher one on PLG and PLR (one run each). The passivity term is a safeguard (the output layer already ensures it), and the two smoothness terms remove jumps and ripple with weights small enough to keep real narrow resonances. The last term asks the real and imaginary parts to be consistent with causality [32]: for $S = r + jx$ at angular frequencies ω_k

$$\hat{r}_k = -\sum_{j \neq k} \frac{x_j\, \Delta\omega}{\pi(\omega_j - \omega_k)}, \quad \hat{x}_k = \sum_{j \neq k} \frac{r_j\, \Delta\omega}{\pi(\omega_j - \omega_k)}. \quad (7)$$

The signs in (7) follow the $e^{+j\omega t}$ time convention of the dataset. Because the band is finite, the estimates are approximate, hence the small weight.

Training uses AdamW [33] (learning rate 2×10^{-4}, weight decay 10^{-5}), batches of 256 and gradient clipping at 1; the learning rate follows a cosine from 2×10^{-4} to 2×10^{-6} over 300 epochs, and the network after the last epoch is used, so the test layouts play no part in training or in the choice of any setting. Mirroring a layout about the x- or y-axis keeps r_{xx} and reverses the sign of r_{yx} (both mirrors together keep both). In training, each layout is mirrored at random with this rule. At test time, the network predicts the four mirror images, restores the sign of r_{yx}, and averages magnitudes and unit phasors separately; averaging the complex values instead shrinks the magnitude wherever the four phases differ. Rotations by $90°$ are not used: they exchange x and y. Each network of Table III is trained on one class; 300 epochs on 108,000 layouts took 6.5 h on one NVIDIA RTX PRO 4000 shared with other jobs, and on 27,000 layouts about 1 h.

V. Surrogate Results

Table III has the form of Table I of MSDCNN [13]: each row is trained on one class and tested on each class, with the mean-squared error (MSE) of $|r_{xx}|$ over the 101 frequencies from 2 to 12 GHz and the 3,000 test layouts. On its own class the proposed network has a lower error than MSDCNN, with one architecture for all, whereas MSDCNN needed ResNet34S (21.3 M parameters) for PTN: 0.871×10^{-3} against 2.760×10^{-3} on PTN (68 % lower, 27 % of the parameters), 2.64×10^{-3} against 10.97×10^{-3} for RDN27000 (76 % lower) and 1.26×10^{-3} against 4.24×10^{-3} for RDN108000 (70 % lower). The mean phase error of r_{xx}, weighted by $|r_{xx}|$, is

TABLE II
TERMS OF THE TRAINING LOSS (6), APPLIED TO EACH OUTPUT; $\hat{S}$ IS THE PREDICTION, S THE DATA, k THE FREQUENCY INDEX

Term	What it measures	Formula	Weight
$\mathcal{L}_{\mathrm{reim}}$	error of real and imaginary parts	$\frac{1}{2K}\sum_k \left[(\mathrm{Re}\,\hat{S}_k - \mathrm{Re}\,S_k)^2 + (\mathrm{Im}\,\hat{S}_k - \mathrm{Im}\,S_k)^2\right]$	1.00
$\mathcal{L}_{\mathrm{dB}}$	magnitude error in decibels	$\frac{1}{K}\sum_k \left(20\log_{10}\max(\lvert\hat{S}_k\rvert,\epsilon) - 20\log_{10}\max(\lvert S_k\rvert,\epsilon)\right)^2$, $\epsilon = 3\times10^{-4}$ (r_{xx}), 10^{-2} (r_{yx})	0.50
$\mathcal{L}_{\mathrm{lin}}$	magnitude error in linear scale	$\frac{1}{K}\sum_k \left(\lvert\hat{S}_k\rvert - \lvert S_k\rvert\right)^2$	0.30
$\mathcal{L}_{\mathrm{ph}}$	phase error (no 2π jumps)	$\sum_k w_k \lvert\hat{S}_k/\lvert\hat{S}_k\rvert - S_k/\lvert S_k\rvert\rvert^2 / \sum_k w_k$, $w_k = 1$ (r_{xx}), $\min(1, \lvert S_k\rvert/0.1)$ (r_{yx})	0.30
$\mathcal{L}_{\mathrm{pass}}$	total reflection above one	$\frac{1}{K}\sum_k \max\left(0, \sqrt{\lvert\hat{S}_{xx,k}\rvert^2 + \lvert\hat{S}_{yx,k}\rvert^2} - 1\right)^2$ (once, not per output)	0.08
$\mathcal{L}_{\mathrm{TV}}$	frequency-to-frequency jumps	$\frac{1}{K-1}\sum_k \lvert\hat{S}_{k+1} - \hat{S}_k\rvert$	0.06
$\mathcal{L}_{\mathrm{curv}}$	ripple (curvature)	$\frac{1}{K-2}\sum_k \lvert\hat{S}_{k+1} - 2\hat{S}_k + \hat{S}_{k-1}\rvert^2$	0.02
$\mathcal{L}_{\mathrm{KK}}$	Kramers–Kronig consistency	$\frac{1}{K}\sum_k \left[(\mathrm{Re}\,\hat{S}_k - \hat{r}_k)^2 + (\mathrm{Im}\,\hat{S}_k - \hat{x}_k)^2\right]$, estimates (7)	0.02

TABLE III
TEST MSE OF $|r_{xx}|$ ($\times10^{-3}$, 2–12 GHz), TRAINED ON ONE CLASS (ROWS), TESTED ON EACH CLASS (COLUMNS)

	Network	Params	PLG	PLR	PTN	RDN	$\lvert r_{yx}\rvert$
MSDCNN	PLG27000 (Res18S)	11.2 M	–[a]	–	7.517	17.436	–
	PTN27000 (Res18S)	11.2 M	–[a]	–	4.382	12.409	–
	PTN27000 (Res34S)	21.3 M	–[a]	–	2.760	17.377	–
	RDN27000 (Res18S)	11.2 M	–[a]	–	5.257	10.971	–
	RDN108000 (Res18S)	11.2 M	–[a]	–	13.631	4.242	–
Edge-aware	PLG27000	1.88 M	0.145	13.0	25.7	14.0	–
	PLR27000	1.88 M	0.647	0.193	21.1	24.3	–
	PTN27000	1.88 M	9.16	15.0	3.96	14.0	–
	RDN27000	1.88 M	21.1	13.6	4.89	13.2	–
	RDN108000	1.88 M	5.97	7.67	3.62	9.43	–
Proposed	PLG27000	5.83 M	**0.084**	8.08	6.20	15.0	0.102
	PLR27000	5.83 M	0.479	**0.062**	20.3	29.9	0.079
	PTN27000	5.83 M	0.520	0.944	0.871	5.79	0.833
	RDN27000	5.83 M	2.11	1.53	1.09	2.64	2.87
	RDN108000	5.83 M	0.930	0.495	**0.522**	**1.26**	1.05

3,000 test layouts per class; 101 frequencies (MSDCNN: 32). MSDCNN from [13, Table I] (its own random splits: 3,000 test layouts for PLG, 300 for PTN and RDN), parameters from [14, Table 2]; edge-aware: reproduction of [18]. [a]MSDCNN reports PLG over 9.5–12 GHz only. $|r_{yx}|$: test MSE of the cross-polarized reflection on the network's own class. Bold: lowest error on each class.

1.2–5.6° on the diagonal. Off the diagonal, networks trained on polygons fail on other classes, as for MSDCNN, while RDN108000 predicts PLG, PLR and PTN with 0.93×10^{-3}, 0.50×10^{-3} and 0.52×10^{-3}, on PTN below the PTN network itself: random layouts contain every kind of shape, and with graph attention the network learns responses that carry over. The same networks predict $|r_{yx}|$ with MSE of 0.08×10^{-3} (PLR) to 2.9×10^{-3} (RDN27000) on their own class, 1.05×10^{-3} for RDN108000, with a weighted phase error of 7.8°. Fig. 3 compares predicted and simulated spectra of each class network, as in Fig. 9 of MSDCNN.

The edge-aware surrogate [18] has the same frequency cross-attention, GRU and polar output layer, but encodes the layout with a shallow CNN (3×3 convolution, two convolution and max-pooling blocks, a final convolution; 1.88 M parameters). Reproduced from its description, including the $4\times$ weighting of the band edges in its loss, and trained like the proposed network, it works on polygons (0.145×10^{-3} on PLG, 0.193×10^{-3} on PLR) but fails on PTN and RDN (3.96×10^{-3} and 13.2×10^{-3}, coefficients of determination R^2 of 0.01 and -0.05, i.e., barely better than predicting the mean spectrum); 108,000 RDN layouts lower its error only to 9.43×10^{-3}. Four local convolutions before pooling suffice for one filled shape, not for how metal connects across a random cell. With the residual backbone and graph attention in place of the shallow encoder, the proposed network is 1.7–3.1 times more accurate on PLG and PLR and 4.5–7.5 times on PTN and RDN.

Trained on 27,000 layouts without the PGA blocks, the network has a higher test error: 0.935×10^{-3} instead of 0.871×10^{-3} for $|r_{xx}|$ and 0.926×10^{-3} instead of 0.833×10^{-3} for $|r_{yx}|$ on PTN, and 2.80×10^{-3} instead of 2.64×10^{-3} and 3.06×10^{-3} instead of 2.87×10^{-3} on RDN (one run each): the nine PGA blocks lower the error by 5–10 % with 1.1 % of the parameters. The network predicts a layout in about 0.13 ms on the GPU in batches of 2048, against about 10 min for one CST simulation.

VI. A VAE DESIGN SPACE FOR RANDOM LAYOUTS

A particle swarm moves particles through a continuous space, whereas a layout is binary: working on the 256 pixels directly, every position must be rounded before scoring. A variational autoencoder (VAE) [11], [35] offers a continuous alternative: its encoder maps a layout to a latent vector z of d numbers, its decoder maps any z back to a layout, and every point the swarm visits decodes to a valid layout. The VAE is trained on RDN layouts, which can represent any layout.

The encoder has a 3×3 convolution to 32 channels at 16×16 and two stride-2 3×3 convolutions to 64 channels at 8×8 and 128 at 4×4, each with batch normalization and LeakyReLU; two linear layers give the mean $\boldsymbol{\mu}$ and log-variance $\log\boldsymbol{\sigma}^2$, and $z = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$. The decoder mirrors it (linear layer to $128 \times 4 \times 4$, two transposed convolutions, a final convolution giving each pixel's probability of copper; above 0.5 is copper). The loss is

$$\mathcal{L}_{\mathrm{VAE}} = 2\,\mathrm{BCE} + \beta\,\mathrm{KL}, \tag{8}$$

the binary cross-entropy (BCE) averaged over the 256 pixels and the Kullback–Leibler divergence (KL) from $\mathcal{N}(0, I)$ averaged over the d latent values, with $\beta = 0$ for the first five epochs and a half-cosine increase to $\beta = 1$ over the next 30 [36]. The encoder blocks use 3×3 convolutions and the decoder 4×4 transposed convolutions of stride 2, with LeakyReLU (slope 0.2). Adam [34] with a learning rate decaying exponentially from 10^{-3} (towards 10^{-5} at epoch 600), batches of 512, one random rotation or mirror per batch, and early stopping after 30 epochs without improvement of the loss on the 3,000 RDN test layouts (counted from epoch 35) are used; a latent value is counted as informative when its mean KL exceeds 0.01.

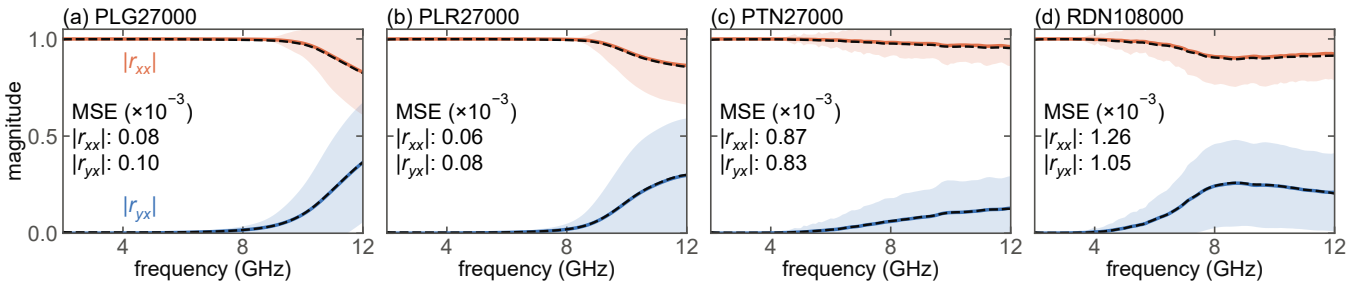


Fig. 3. $|r_{xx}|$ and $|r_{yx}|$ over the 3,000 test layouts of each class: mean (solid) and ± 1 standard deviation (band) of the CST data, and mean prediction of the class network (dashed; RDN: RDN108000), with its test MSE.

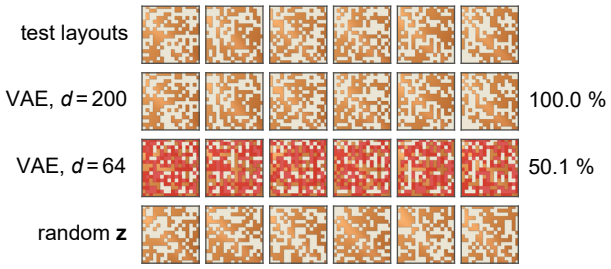


Fig. 4. RDN test layouts, their reconstructions by the VAE with $d = 200$ and $d = 64$ (wrong pixels in red; right: share of correct pixels over the 3,000 test layouts), and layouts decoded from random z ($d = 200$).

The latent size follows from the information in a layout. Every RDN pixel is an independent coin flip, so a layout carries 256 bits. Each unit of information stored in z costs at least one unit of KL and saves at most one of BCE [37]; with the averaging in (8), the KL weight per layout is $\beta_{\text{eff}} = \beta N/(2d)$ with N pixels. For $\beta_{\text{eff}} > 1$ storing information costs more than it saves, and the VAE ignores its input (posterior collapse [36]). With $N = 256$, $d = 64$ gives $\beta_{\text{eff}} = 2$, and such a VAE reconstructs 50.1 % of the pixels (chance), with no informative latent value; d must exceed 128, and $d = 200$ ($\beta_{\text{eff}} = 0.64$, stopped at epoch 79) reconstructs all test pixels, with 192 of the 200 latent values informative. Layouts decoded from random $z \sim \mathcal{N}(0, I)$ are random patterns with 53 % copper (dataset: 50 %), i.e., in the region where the surrogate was trained (Fig. 4).

VII. Design Method

A. Objectives

Each design is scored by three objectives (Table IV): a bandwidth B, a quality of the response in that band, and the number D of corner contacts, 2×2 windows in which only one diagonal pair of pixels is copper, plus the number of isolated pixels, copper pixels with no copper neighbour along an edge. A corner contact has zero width, so whether it conducts depends on etching tolerance and, in simulation, on meshing, and an isolated pixel is easily lost; a typical RDN layout has $D \approx 38$. The bandwidths are fractional, $2(f_{\text{hi}} - f_{\text{lo}})/(f_{\text{hi}} + f_{\text{lo}})$, with band edges interpolated linearly between frequencies (in decibels for the coding pair); band qualities are averages over the frequencies inside the band.

The LP-to-CP converter (10 GHz) reflects a linearly polarized wave as a circularly polarized one, which requires $|r_{xx}| = |r_{yx}|$ with a 90° phase difference. With the Stokes parameters $S_0 = |r_{xx}|^2 + |r_{yx}|^2$ and $S_3 = 2\,\text{Im}(r_{xx}^* r_{yx})$, the axial ratio is $\text{AR} = \cot\chi$ with $\sin 2\chi = |S_3|/S_0$. B is the 3-dB AR band containing $f_0 = 10$ GHz, the quality is the mean AR in it, and the lowest reflected power $10\log_{10} S_0$ in the band, L, is a constraint ($L \geq -0.5$ dB).

TABLE IV
Applications and Objectives

Device	B (%), maximize	Quality	Target
LP-to-CP, 10 GHz	3-dB AR band	mean AR (min.)	$B \geq 10$ %, $L \geq -0.5$ dB, $D \leq 5$
Rotator, 10.2 GHz	PCR > 0.9 band	mean $\lvert r_{yx}\rvert^2$ (max.)	$B \geq 5$ %, $\lvert r_{yx}\rvert^2 \geq 0.8$, $D \leq 5$
Coding pair	$s < -10$ dB band, both pol.	mean reduction (max.)	–
Penalties	$(30 - B)/30$; $\overline{\text{AR}}/3$ dB or $1 - \overline{\lvert r_{yx}\rvert^2}$; $D/60$. Coding: $(80 - B)/80$, $(25 - R)/25$ dB, $D/120$. Extra penalty if f_0 is outside the band: $(\text{AR}(f_0) - 2.5)/20$, $0.93 - \text{PCR}(f_0)$; coding, if no band: $(11 - R_{\max})/11$; for CP also $(-0.5 - L)/3$ if $L < -0.5$ dB. Without a band, the quality is taken at f_0 (coding: largest reduction).		
Search margins	B symmetric about f_0 (10); thresholds AR < 2.5 dB, PCR > 0.93, $s < -11$ dB.		

For the 90° polarization rotator (10.2 GHz), the polarization conversion ratio is $\text{PCR} = |r_{yx}|^2/(|r_{xx}|^2 + |r_{yx}|^2)$; B is the band of PCR > 0.9 containing f_0 and the quality the mean $|r_{yx}|^2$ in it. Broadband rotation in this cell family ends at 11.0–11.25 GHz (FDTD of designs from a search at 10.9 GHz), so $f_0 = 10.2$ GHz is placed at the centre of the rotation band.

For the 1-bit coding checkerboard, two cells A and B in a checkerboard of $N \times N$-cell tiles reflect, in the specular direction and when the tiles are large enough, the area average of their reflections [2], [41]. Relative to a metal plate, the specular reflectance for x-polarized incidence is

$$s_x = \tfrac{1}{4}\left(|r_{xx}^A + r_{xx}^B|^2 + |r_{yx}^A + r_{yx}^B|^2\right), \tag{9}$$

and s_y follows from the same formula with the transposed layouts, since exchanging x and y turns r_{yy} of a layout into r_{xx} of its transpose. The objectives use $s = \max(s_x, s_y)$, so that the reduction holds for both polarizations: B is the widest band with $s < -10$ dB (anywhere in 2–12 GHz), the quality is the mean reduction $-10\log_{10} s$ in it, capped at 30 dB per frequency, and D counts both cells. Three pair structures are searched: a free pair (512 pixels, or two VAE vectors); a mirror pair, $B = \text{mirror}(A)$, whose cross-polarized reflections cancel exactly because a mirror reverses the sign of r_{yx} (a polarization-conversion checkerboard [42]); and a diagonal mirror pair, with A symmetric about its diagonal (136 free pixels), for which $s_x = s_y$.

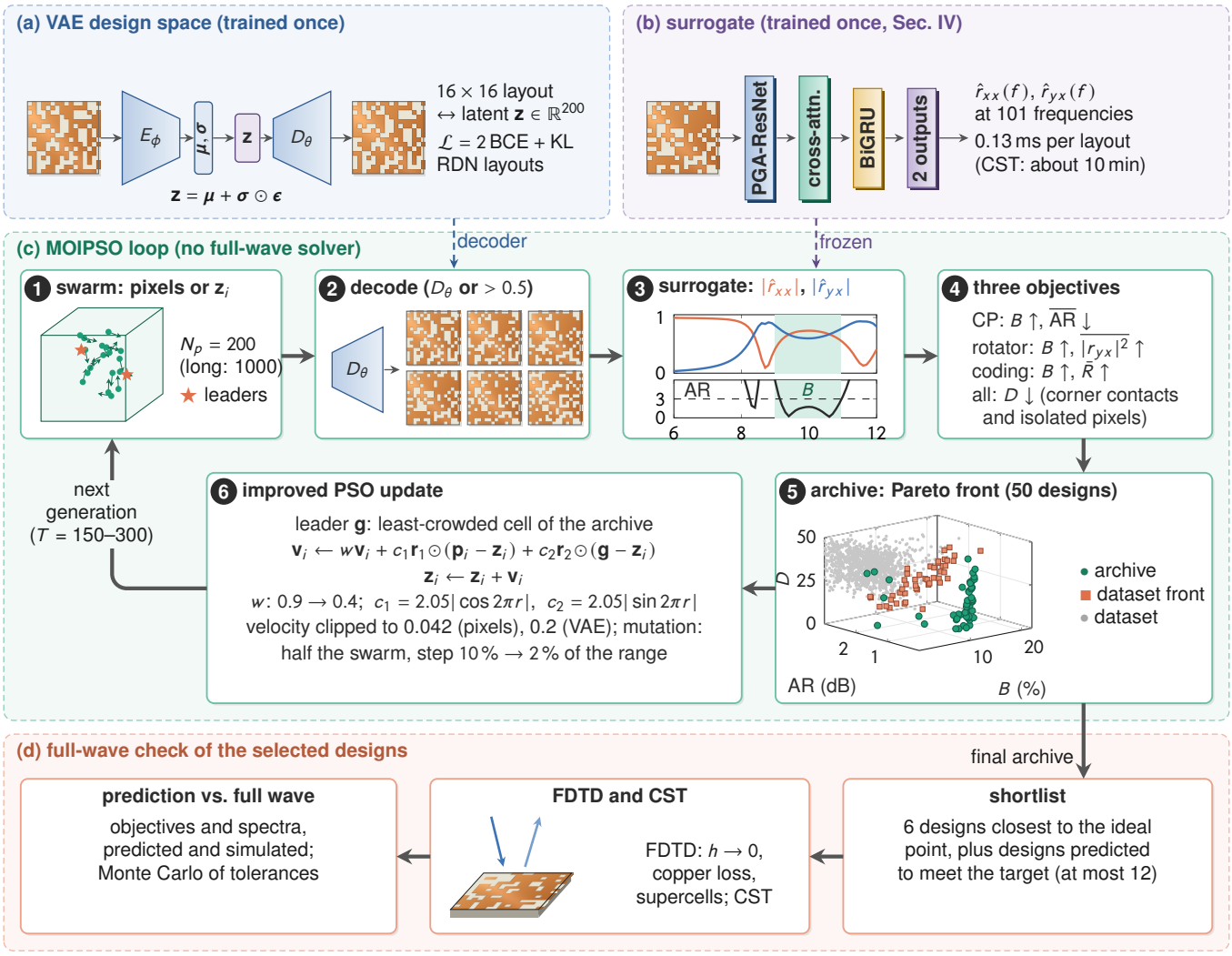


Fig. 5. Design method. (a) VAE of RDN layouts and (b) surrogate, both trained once. (c) One generation of MOIPSO, steps 1–6: the particles (pixels, or VAE vectors decoded by the VAE) become layouts, the surrogate predicts r_{xx} and r_{yx} (example: the best LP-to-CP design, with its axial ratio and band B), the three objectives are computed, the archive (drawn over the RDN dataset for the LP-to-CP converter) keeps the Pareto front, and the PSO update moves the particles. (d) FDTD check of the selected designs.

Scored as "the band that contains f_0", the bandwidth lets the optimizer place f_0 at a band edge, where a shift of the whole response by 1–2 %, the size of meshing, permittivity or etching tolerances, moves f_0 out of the band. During the search, the bandwidth is therefore scored symmetrically about f_0,

$$B_{\mathrm{sym}} = 2\min(f_0 - f_{\mathrm{lo}},\ f_{\mathrm{hi}} - f_0)/f_0, \tag{10}$$

and the band thresholds carry a margin (AR < 2.5 dB, PCR > 0.93, $s < -11$ dB) against the surrogate's error. All reported values use the standard definitions of Table IV.

B. Multi-Objective Particle Swarm

Each objective is mapped to a penalty between 0 (ideal) and 1 (Table IV); a design whose f_0 lies outside the band receives an extra penalty on all three, which guides the swarm to the band condition. Design a dominates design b if it is at least as good in every objective and better in one; the non-dominated designs form the Pareto front. Fronts are compared by the hypervolume (HV) [38]: with penalty vectors $\mathbf{p}$ and reference point $\mathbf{1}$,

$$\mathrm{HV}(P) = \mathrm{vol}\Big(\bigcup_{\mathbf{p}\in P} [p_1, 1] \times [p_2, 1] \times [p_3, 1]\Big), \tag{11}$$

between 0 and 1, larger for fronts that are better, wider or both. It is computed by Monte Carlo integration with 200,000 uniform points (statistical error below 0.002; 20,000 points, error below 0.004, during the searches). The fronts of the searches (surrogate) and of the dataset (its CST spectra) use the same penalties, so their HVs compare directly.

The optimizer is adapted from the improved multi-objective particle swarm optimizer (MOIPSO) of [40], whose acceleration coefficients and half-swarm Gaussian mutation it keeps; its archive and leader selection follow the grid-based MOPSO [39], and the inertia schedule, the floor of the mutation step and the velocity clip are chosen here (Table V). Each particle i has a position z_i and a velocity v_i, updated as

$$v_i \leftarrow \omega v_i + c_1 \mathbf{r}_1 \odot (p_i - z_i) + c_2 \mathbf{r}_2 \odot (g_i - z_i), \quad z_i \leftarrow z_i + v_i, \tag{12}$$

TABLE V
MOIPSO SETTINGS AND SEARCHES

Setting	Value
Search spaces	pixels $[0,1]^{256}$ (threshold 0.5); VAE $z \in [-1.5, 1.5]^{200}$
Inertia; acceleration	0.9→0.4 linear; $2.05\lvert\cos 2\pi\rho_1\rvert$, $2.05\lvert\sin 2\pi\rho_2\rvert$
Archive	50 designs, grid of 7^3 boxes, crowding-based leaders
Mutation; velocity	half the swarm, step 10 %→2 % of range; ± 0.042 (pixels), ± 0.2 (VAE)
Searches per device (CP, rotator)	pixel and VAE: 3×(200 particles, 150 gen.) and 1000×300 each
Searches (coding)	diagonal mirror (136 px), mirror (256 px; VAE 200) 1000×300; free pair (512 px; VAE 400) 600×250
Selection	CP, rotator: 6 designs of the best-HV run with the smallest penalty sum, then designs of all runs predicted to meet the search target, by penalty sum, at most 12. Coding: non-dominated pairs of all runs; 3 widest bands with $D \leq 10$, then smallest penalty sum, 6 pairs

with uniform random vectors $\mathbf{r}_1, \mathbf{r}_2$, the particle's best position p_i, a leader g_i from the archive, inertia ω decreasing from 0.9 to 0.4, and $c_1 = 2.05\lvert\cos 2\pi\rho_1\rvert$, $c_2 = 2.05\lvert\sin 2\pi\rho_2\rvert$ with random ρ_1, ρ_2 per generation. After each move, each velocity component is clipped to ± 0.042 in pixel space and ± 0.2 in VAE space. An external archive keeps up to 50 non-dominated designs, one entry per layout; leaders are drawn from grid boxes in objective space (7 divisions per objective) with probability inversely proportional to their occupancy [39], and when the archive is full the members with the smallest crowding distance are removed [7]. Half of the swarm receives a Gaussian mutation whose step falls from 10 % to 2 % of the range over the run; particles that leave the box are reflected. p_i is replaced when the new position dominates it, and with probability 0.5 when neither dominates. In pixel space, positions lie in $[0,1]^{256}$ and a pixel is copper above 0.5; in VAE space, in $[-1.5, 1.5]^{200}$ with $z \sim \mathcal{N}(0, I)$, clipped to the box, at the start. One generation decodes the positions, predicts the reflection of all layouts in one batch (with the four-mirror average; for coding, of A, B and their transposes), computes the objectives and updates archive, bests and particles; no full-wave simulation is used.

VIII. FULL-WAVE VERIFICATION

The selected designs are simulated with an FDTD solver written for this cell, so that every number can be reproduced without a commercial solver.

A Yee grid [43], [44] of cubic cells of size h (0.25 and 0.125 mm, i.e., 2 and 4 cells per pixel) covers one period with periodic boundaries in x and y at normal incidence; the time step is $0.99h/(c\sqrt{3})$. The substrate loss is a conductivity $2\pi f_\ell \varepsilon_0 \varepsilon_r \tan\delta$, exact at $f_\ell = 10$ GHz, and tangential fields on the air–substrate plane use the averaged permittivity. All copper is lossy, with the surface resistance of copper at 10 GHz, $R_s = \sqrt{\pi f \mu_0/\sigma} = 0.026\,\Omega$ ($\sigma = 5.8 \times 10^7$ S/m; both copper layers are much thicker than the 0.66 µm skin depth), held constant over frequency: the pixels are zero-thickness resistive sheets on the substrate, updated with exponential time stepping, and the ground is a surface-impedance boundary, $E_x = -R_s H_y$, $E_y = R_s H_x$, with H half a cell above it. Tangential electric field on every grid edge in the closure of a copper pixel is set by the sheet, so two pixels touching at a corner conduct; with cut corners the dataset is matched worse (1 of 24 cells closer). The top is a uniaxial perfectly matched layer (PML) graded in z (Gedney form [45], 12 cells, cubic grading, $R_0 = 10^{-8}$) backed by a conductor; for the large checkerboard cells a convolutional PML [46] (20 cells, $\kappa_{\max} = 5$, α at 3 GHz) is used, because the UPML grows slowly in time there. A Gaussian-modulated pulse (centre 7 GHz, bandwidth 2.33 GHz) is launched 5 mm above the cell; the reflected field is the plane average of E_x and E_y 2 mm above the source (the plane average over a period is exactly the specular order) minus the incident field of a one-dimensional FDTD with the same h and time step, and runs stop when the late-time signal falls below 10^{-4} of its peak (30 ns at most). The reflection is moved to the cell surface with the numerical wavenumber of the grid and then to the dataset's port plane, $\Delta = 10.705$ mm above the cell, the distance found by fitting the CST phase of the empty cell to the analytic grounded slab [48] (FDTD gives 10.69 mm; the difference is below 0.5° in phase).

Grid dispersion shifts resonances down by about 2.5 % at $h = 0.25$ mm and 0.4–0.6 % at 0.125 mm. Writing $S_h(f) = S_0(f(1+\delta_h))$ with $\delta_h \propto h^2$, the two grids give $\delta_{0.125} = (s-1)/3$ from the scale s that best aligns the two spectra (least MSE of $|r_{xx}|$ and $|r_{yx}|$, s from 0.97 to 1.08 in steps of 5×10^{-4}, real and imaginary parts interpolated linearly), and the converged spectrum $S_0(f) = S_{0.125}(f/(1+\delta_{0.125}))$ (a Richardson extrapolation [47] of the frequency axis).

The solver is validated against the dataset. The empty cell matches the analytic grounded slab with the same ground resistance within $|\Delta r| = 6\times10^{-4}$. On 96 dataset cells (24 per class) at $h = 0.25$ mm, the MSE of $|r_{xx}|$ and $|r_{yx}|$ against CST is 1.11×10^{-3}, and the lowest reflected power $|r_{xx}|^2 + |r_{yx}|^2$ differs from CST by a median of 0.8×10^{-3} (RDN cells: 5×10^{-3}). On resonant PLG and PLR cells, the extrapolation to $h \to 0$ lowers the MSE from 0.3–0.9×10^{-3} at 0.125 mm to 0.02–0.17×10^{-3}. On the four RDN cells of the dataset that rotate polarization over 9.8–10.6 GHz (PCR > 0.9; all four are training cells), the mean $|r_{yx}|^2$ over 9.5–10.9 GHz in FDTD differs from CST by -0.017, and that of RDN108000 by -0.033.

The selected designs are also simulated in CST Studio Suite, with the geometry and materials of Section II, unit-cell boundaries, Floquet ports for both polarizations at the dataset's port plane and the frequency-domain solver from 2 to 12 GHz; r_{xx} and r_{yx} are read from the two Floquet modes, and one cell takes about 10 min.

A periodic checkerboard is simulated as one supercell of 2×2 tiles of $N \times N$ cells, each cell with its own margin, in the same way (for y-polarized incidence, the transposed board), at $h = 0.25$ mm only. Diffracted orders leave through the PML; the plane average is the specular order, and the reduction is the specular reflected power relative to a metal plate. Because large supercells still show a slow late-time growth, the reflected signal is taken over the first 15 ns with a Hann taper over its last quarter; on stable runs this matches 40 ns runs within 0.1–0.3 dB. The frequency axis is corrected with $\delta_{0.25} = 4(s-1)/3$, s the median scale of the two unit cells.

Manufacturing tolerances are estimated for the best design

TABLE VI
DESIGN RESULTS (FDTD EXTRAPOLATED TO $h \to 0$, AND CST)

	LP-to-CP, 10 GHz	Rotator, 10.2 GHz	Coding pair
Dataset: best B	20.9 % ($D = 38$)	12.7 % ($D = 45$)	13.1 %[a] ($D \geq 24$)
Dataset: meet target	0 of 111,000	0 of 111,000	–
HV, dataset / search[b]	0.175 / 0.486	0.149 / 0.473	0.093[a] / 0.190
Meet target, FDTD / CST	12 / 12 of 12	12 / 12 of 12	–
B, FDTD	13.2–17.6 %	13.9–16.1 %	16.0 %[c]
B, CST	13.9–18.4 %	14.5–16.8 %	18.4 %[d]
Quality, FDTD	AR 0.86–1.44 dB	$\lvert r_{yx}\rvert^2$ 0.90–0.92	13.7 dB mean[c]
f_0 to nearest edge	6.4–8.4 %	6.7–8.2 %	–
D	0–4	0–5	0 (pair)

[a] Best pair of the 111,000 cells and their mirror images, x-polarization only, from (9). [b] Predicted, search objectives (10). [c] Periodic checkerboard, 6×6-cell tiles, both polarizations (FDTD). [d] Unit cells with (9), both polarizations (CST).

of each device by a Monte Carlo simulation of 64 boards: ε_r uniform in 2.60–2.70, $\tan\delta$ in 0.002–0.004 and the substrate thickness in 0.95–1.05 mm, and every corner contact opened independently with probability 0.5 (the same contacts of a cell for both polarizations), simulated at $h = 0.125$ mm with the measured mesh shift. The thickness is resolved by stretching the substrate cells in z, which matches the analytic grounded slab within $7{\times}10^{-4}$ for 0.95 and 1.05 mm; with nominal values the 0.125 mm grid reproduces the extrapolated spectra of the designs.

IX. DESIGN RESULTS

A. Dataset Baseline and Searches

The 111,000 RDN layouts with their CST spectra are the baseline (Table VI). No layout meets the LP-to-CP or the rotator target: the dataset's widest 3-dB AR band around 10 GHz is 20.9 %, with $D = 38$, and every layout with at least 10 % has $D \geq 20$; the widest PCR band around 10.2 GHz is 12.7 %, with $D = 45$, and every layout with at least 5 % has $D \geq 27$. For the coding pair, the baseline is the best pair a designer could pick from the library of 111,000 cells and their mirror images (whose response follows exactly): among 2.5×10^{10} pairs, the 16 partners nearest to $-A$ in four frequency windows were scored exactly for every cell A (7.1 M candidates), giving at most 13.1 % at -10 dB for x-polarization alone (the dataset has no y-incidence), with $D \geq 24$ per pair. With the four-mirror average, a standard search scores 30,200 layouts in about 30 s and a long search 301,000 layouts in about 3 min on one GPU; one CST simulation takes about 10 min. All 16 converter and rotator searches pass the HV of the dataset, after 1,000–20,000 scored layouts, i.e., 1–18 % of the number of dataset layouts, and the best searches end at HV 0.486 against 0.175 for the converter and 0.473 against 0.149 for the rotator; the searched fronts (Fig. 6) dominate the dataset fronts.

With the standard budget the VAE space gives the better fronts: HV 0.41–0.46 against 0.32–0.35 in pixel space for the converter and 0.44–0.45 against 0.34–0.39 for the rotator, with 2–8 archived designs predicted to meet the target per run against at most 3. With the long budget both spaces reach about the same HV (converter 0.479 VAE, 0.486 pixel; rotator 0.473 VAE, 0.458 pixel). For the coding pair, the diagonal pixel search gives the best front (HV 0.190; mirror pair 0.183 in pixel and 0.179 in VAE space), and among free pairs the VAE space is far ahead (0.133 against 0.071).

B. Polarization Converter and Rotator

All 12 selected converters meet the target in FDTD and in CST (Table VI): 3-dB AR bandwidths of 13.2–17.6 % (FDTD) and 13.9–18.4 % (CST) around 10 GHz, mean AR of 0.86–1.44 dB (FDTD; predicted 0.82–1.24 dB), lowest in-band reflection of -0.18 to -0.23 dB (FDTD) and -0.11 to -0.13 dB (CST), and D of 0–4. 10 GHz lies 6.4–8.4 % from the nearest band edge. The predicted band edges agree with the extrapolated FDTD within 0.04 GHz (lower) and 0.11 GHz (upper), and with CST within 0.04 and 0.10 GHz (Fig. 7).

All 12 selected rotators meet the target in FDTD and in CST: PCR > 0.9 bandwidths of 13.9–16.1 % (FDTD) and 14.5–16.8 % (CST) around 10.2 GHz with 6.7–8.2 % from f_0 to the nearest edge, mean $\lvert r_{yx}\rvert^2$ of 0.90–0.92 (FDTD) and 0.92–0.94 (CST), against a prediction of 0.85–0.87, and D of 0–5; the band edges agree within 0.13 GHz. The under-prediction of $\lvert r_{yx}\rvert^2$ is also seen on the four dataset cells that rotate polarization near 10 GHz (Section VIII), which are rare in the training data: there the surrogate is 0.033 low, while FDTD matches CST within 0.017.

C. Coding Checkerboard

The diagonal and plain mirror searches reach the widest predicted bands (18.5 and 18.1 % at -11 dB for both polarizations), the free pairs 14.0 %. For the best diagonal mirror pair, with no corner contacts or isolated pixels, the two unit cells with (9) give 17.6 % at -10 dB in FDTD and 18.4 % in CST (surrogate: 18.8 %). The full-wave simulation of the periodic checkerboard shows that this holds only for large tiles (Fig. 7(c)): between the two resonances of the pair, cells next to the other cell type couple to it and lose part of their polarization conversion, and the lowest reduction over 9.5–10.8 GHz rises from 5.7 dB with 3×3-cell tiles to 7.0, 10.0 and 11.0 dB with 4×4, 6×6 and 8×8-cell tiles. With 6×6-cell tiles (60 mm, period 120 mm) the board reduces the specular reflection by more than 10 dB from 9.31 to 10.93 GHz (16.0 %) for both polarizations, by 13.7 dB on average, and with 8×8-cell tiles from 9.21 to 10.94 GHz (17.2 %, 14.6 dB); with 3×3 and 4×4-cell tiles the widest band is 5.7 and 4.9 %. The averaging model (9) is therefore a good design objective for these resonant cells only with large tiles, and a coding design has to be confirmed by a full-wave simulation of the checkerboard, not of the unit cells alone.

Manufacturing tolerances (Section VIII) leave the converter with the widest band within its target on all 64 Monte Carlo boards, with 3-dB AR bandwidths of 16.2–19.7 % (5th to 95th percentile; nominal 17.6 %). The rotator with the widest band (16.1 %) has five corner contacts and isolated pixels and keeps its target on only 34 of the 64 boards: when some of its corner contacts open, its PCR band breaks up, whereas the substrate tolerances explain little of the spread. The rotator without any corner contacts or isolated pixels (15.5 % in FDTD) meets its target on all 64 boards, with PCR > 0.9 bandwidths of 15.0–15.9 %, and is the one shown in Fig. 7(b): the count D is

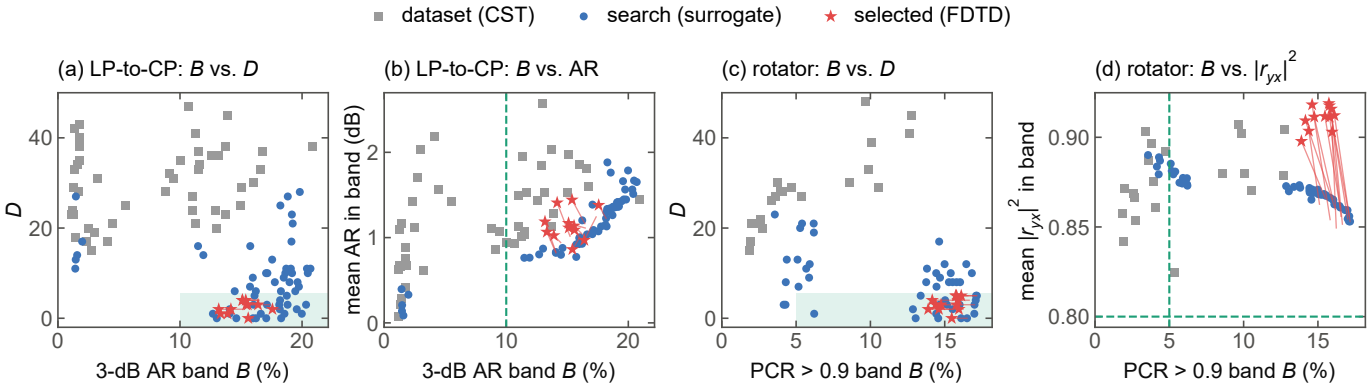


Fig. 6. Pareto fronts of the LP-to-CP converter (a, b) and the rotator (c, d): bandwidth against D (corner contacts and isolated pixels) and against polarization quality. Squares: dataset front (CST spectra); dots: search front (surrogate); stars: selected designs in FDTD ($h \to 0$), with lines from their predicted values. Shaded or dashed: target.

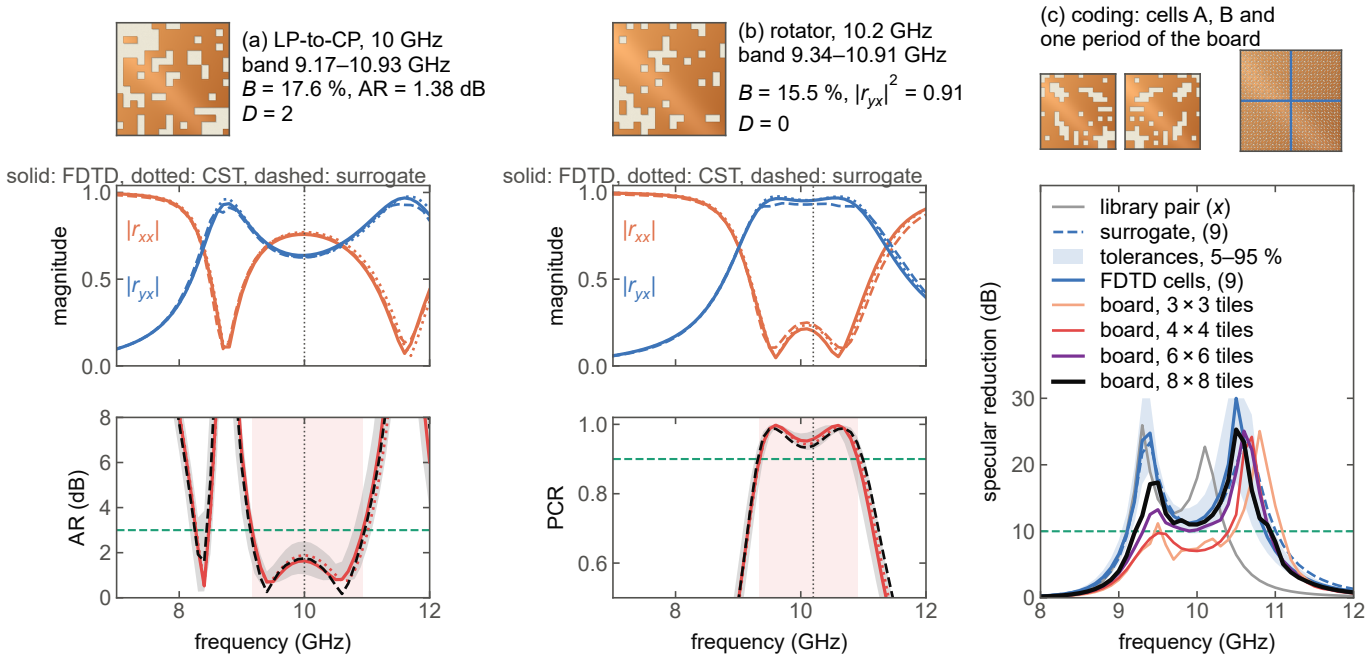


Fig. 7. Best designs. (a) LP-to-CP converter with the widest band and (b) rotator without corner contacts or isolated pixels: layout (16×16 pixels of 0.5 mm, incident field along the rows), $|r_{xx}|$ and $|r_{yx}|$, and axial ratio or PCR; FDTD at $h \to 0$ (solid), CST (dotted) and surrogate (dashed); shaded: band in FDTD; dotted vertical: f_0. (c) Coding checkerboard of a cell and its mirror image: reduction of the specular reflection against a metal plate from (9) with the surrogate and with FDTD of the two cells, and from FDTD of the periodic board with 3×3, 4×4, 6×6 and 8×8-cell tiles (worse of the two polarizations); grey: best library pair (x-polarization only). Shaded grey (a, b) and light blue (c): 5th to 95th percentile over 64 boards with manufacturing tolerances.

worth minimizing for fabrication. The coding pair, also free of them, keeps a -10 dB band of at least 10 % on 81 % of the boards (8.3–18.0 %).

X. Discussion

The optimizer selects the best of about 10^5 predictions, so errors of the surrogate are most likely to appear in the selected designs. On the 24 selected converters and rotators, the MSE of $|r_{xx}|$ and $|r_{yx}|$ between surrogate and CST over 2–12 GHz is 0.13–0.50×10^{-3}, and between surrogate and FDTD 0.18–1.0×10^{-3}, below the RDN test error of the surrogate against CST (1.26×10^{-3} and 1.05×10^{-3}), so the optimizer has not driven the designs into regions where the surrogate is unreliable. Band edges agree within 0.13 GHz, and all 24 designs meet their targets in both solvers. The one systematic error found is conservative: the cross-polarized reflection of the rotators is under-predicted by about 0.05–0.07.

Scoring the bandwidth symmetrically about f_0 gives up part of the band on one side but leaves a margin of 6.4–8.4 % against shifts of the response in frequency.

The unit-cell surrogate cannot see the coupling between different neighbouring cells, so the tile size is part of the design; here it must be at least 6×6 cells.

The surrogate is trained for normal incidence, one substrate and 0.1 GHz frequency spacing. The dataset also holds narrow absorption lines (0.04–0.1 GHz wide) that this spacing does not resolve; absorbers and sensors need a surrogate trained at the full 0.01 GHz resolution.

XI. Conclusion

A surrogate model combining a convolutional backbone with pixel-wise graph attention, frequency cross-attention, a bidirectional GRU and a passive two-output layer predicts the co- and cross-polarized reflection of 16×16 pixel layouts; one architecture serves all four SUTD-PRCM classes and is 68–76 % more accurate than the class-specific MSDCNN models on shape combinations and random layouts. Used in place of the solver, the surrogate lets a multi-objective particle swarm design three devices from random layouts, scored by bandwidth, quality and the number of corner contacts and isolated pixels, with the design frequency kept inside the band. FDTD and CST confirm 12 of 12 LP-to-CP converters (13.2–18.4 % 3-dB AR bandwidth at 10 GHz) and 12 of 12 polarization rotators (13.9–16.8 % at 10.2 GHz), targets no dataset layout meets; all 64 boards of a Monte Carlo study of manufacturing tolerances still meet them for the converter and for a rotator without corner contacts or isolated pixels. A coding checkerboard reduces the specular reflection by more than 10 dB over 16.0 % for both polarizations with tiles of 6×6 cells. Code and designs are available from the authors.

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